

Coseismic fault rupture at the trench axis during the 2011 Tohoku-oki earthquake

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Fault rupture during the 2011 Tohoku-oki earthquake, which generated a huge tsunami, is thought to have propagated to a shallow part of the subduction zone^{1–3}. This observation calls into question conceptual models that assume that the shallow part of the plate boundary interface in a seismogenic subduction zone slips aseismically^{4,5}. However, the available observations of the earthquake and tsunami do not have sufficient resolution near to the subduction trench to determine whether coseismic fault slip extended all the way to the trench axis. Here we use seismic reflection data to image the subduction trench axis seawards of the Tohoku-oki earthquake epicentre. We compare an image of a profile taken in 1999 with one acquired along the same profile 11 days after the earthquake. Before the earthquake, we observe a triangular wedge of sediments at the trench axis. After the earthquake, we observe a deformed upheaval structure in the sedimentary layer that is 3 km long and 350 m thick. We suggest that this remarkable deformation structure formed as a result of compression during coseismic slip on the shallow plate interface, implying that fault rupture during the Tohoku-oki earthquake did reach the sea floor at the trench axis. We conclude that the shallow plate interface at the subduction trench axis can slip seismically.

The most surprising feature of the 2011 Tohoku-oki earthquake is the large slip that extended to near the trench, as shown by numerous coseismic slip-distribution models^{1–3,6–8}. Large coseismic displacements of more than 20 m near the trench can also be seen in seafloor geodetic observations^{9,10}. Furthermore, a differential bathymetry study that mapped changes in the seafloor topography from before to after the earthquake, revealed direct evidence for large coseismic displacements (up to 50 m horizontally) of the overriding plate relative to the incoming plate that reach the trench axis¹¹. None of these studies, however, yielded any data where coseismic fault slip reaching the toe of the subduction thrust interface could be identified. As a result, they could not confirm whether the shallowest part of the plate interface in the Japan Trench was a slip zone during the earthquake.

Studies carried out in other subduction zones where large earthquakes have generated large tsunamis, for instance, in the Cascadia, Sumatra and Nankai subduction zones^{12–15}, have examined the extent of updip rupture propagation reaching near the trench, but no study has reported any unambiguous evidence of actual coseismic fault slip reaching the trench during a megathrust subduction earthquake¹⁶, except for possible evidence for slip reaching the sea floor from seismic and high-resolution bathymetry

studies of the Sumatra prism toe^{13,15}. Here, we present detailed seismic reflection images of the area of large slip obtained after the 2011 Tohoku-oki earthquake during a rapid-response geophysical cruise from 14 to 31 March 2011 by research vessel *Kairei* seawards of the earthquake's epicentre (Fig. 1). We also compare the seismic image from this survey with one obtained in 1999, 12 years before the earthquake, to identify coseismic structural deformation near the trench. Furthermore, we present a high-resolution seismic image obtained from 21 October to 11 November 2011 along a part of the same profile around the trench. As the seismic systems used in 1999 and 2011 were different, we compared the seismic images obtained before and after the earthquake only visually, not quantitatively.

Here, we briefly describe the structural features seawards of the epicentre. Following previous studies^{17,18}, we divide the structure horizontally into five areas: the deep-sea terrace, the upper, middle and lower slope and the trench axis (Fig. 2a). Previous seismic studies and ocean-drilling results show that the upper plate beneath the deep-sea terrace and upper slope consists of an older (Cretaceous to Neogene) continental framework and that incoming oceanic sediment consists of a Neogene hemipelagic mudstone resting on a carbonate- and chert-rich Cretaceous pelagic sequence^{17,18}. Beneath the deep-sea terrace, the subducting oceanic crust can be traced down to 12 km below sea level. Seawards of the deep-sea terrace, the upper slope is characterized by a highly reflective zone above the subducting oceanic crust and normal faults that cut the topmost sediments as well as a part of the acoustic basement, which is interpreted as a Cretaceous unconformity^{17,18} (Fig. 2d). The highly reflective zone, which is about 2 km thick and seems to thin towards the land, consists of a series of reflectors that are subparallel to the subducting oceanic basement. A clear landward-dipping normal fault forming a large offset (~1 km) at the Cretaceous basement is observed at the landward edge of the upper slope (common depth point (CDP) 1,800 in Fig. 2), and similar landward-dipping faults are recognized on the seaward side of this large normal fault (Fig. 2d, small arrows). Similar reflective zones above the subducted oceanic crust are widely observed in subduction seismogenic zones^{18,19}. The reflective zone has been interpreted as a shear zone, possibly with high pore-fluid pressure^{19,20}.

A 15-km-wide wedge-shaped structural block called the frontal prism¹⁵, which is interpreted as a mixture of sedimentary material transported from the upper plate and accreted pelagic and hemipelagic sediments^{17,18}, is imaged beneath the middle and lower slopes (CDP 3,500–5,100; Figs 2a and 3). Its seismic structure

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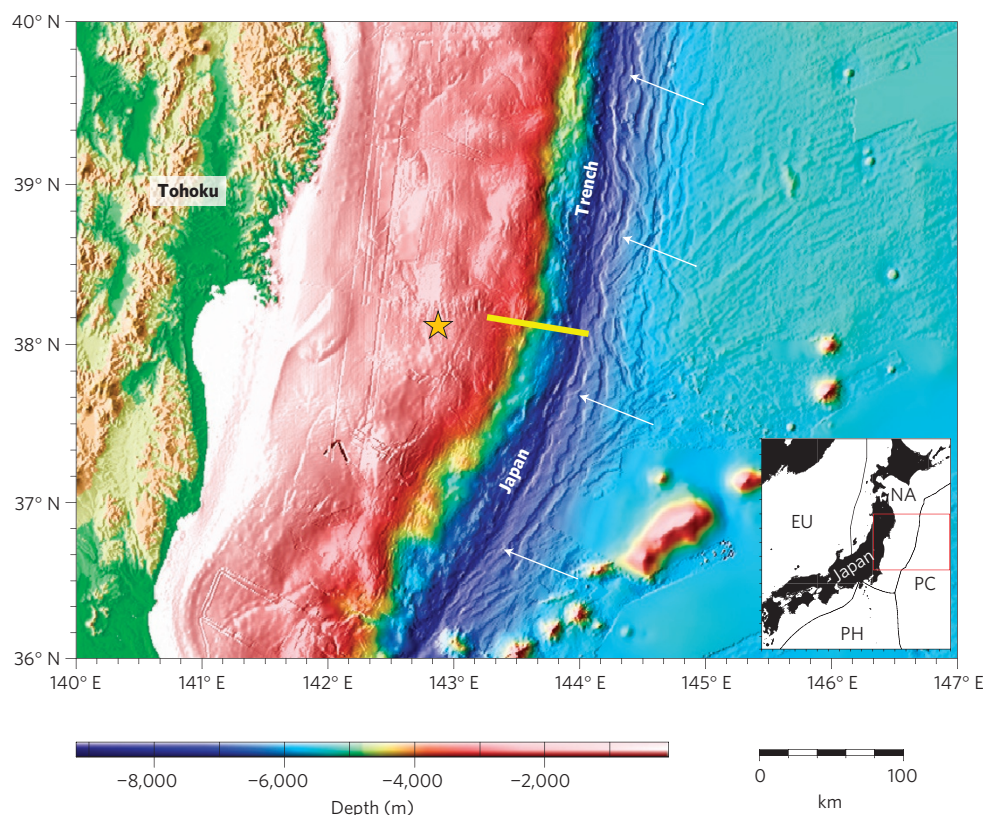


Figure 1 | Map showing the location of the seismic profile. The yellow line and star indicate the profile studied here and the epicentre of the 2011 Tohoku-oki earthquake determined by the Japan Meteorological Agency, respectively. White arrows indicate the relative motion of the Pacific plate, holding the Japan mainland fixed⁶. PC, Pacific plate; NA, North America plate; EU, Eurasia plate; PH, Philippine Sea plate.

is characterized by landward-dipping reflectors (Fig. 3, smaller, shallow arrows) and a subhorizontal reflector above the subducting oceanic basement that is interpreted as the decollement interface (Fig. 3, larger arrows). The landward-dipping reflector seems to terminate at the decollement. Previous studies^{16,20–22} have referred to the most prominent landward-dipping reflector, which lies at the landward edge of the frontal prism here, as a back-stop interface. The seismic character of the seaward half of the frontal prism (CDP 4,500–5,100) beneath the lower slope differs from that of its landward half (CDP 3,500–4,500). The structure of the seaward half is generally transparent, but a weak subhorizontal reflector can be traced in the sediments slightly above the oceanic crust (Fig. 3, larger arrows).

We now present evidence based on bathymetric and seismic reflection data that the Tohoku earthquake rupture reached the trench axis. In our previous study¹¹, differential bathymetry data showed a significant difference close to the trench axis before and after the earthquake (Fig. 2b). Discounting the pair of very large positive (red) and negative (purple) anomalies at the trench axis, which we discuss later, the bathymetric change can be generally divided into two parts: a larger bathymetric deviation is observed seawards of the large normal fault (CDP 1,800) and a smaller deviation is seen landwards of the normal fault. Also, trench-parallel lineation of the deviation can be identified, especially seawards of the normal fault. These characteristics raise the possibility that large coseismic slip may have occurred along the large normal fault²³ or other high-angle landward-dipping reflectors in addition to the 50 m of horizontal coseismic displacement along the megathrust.

To examine this idea, we first transformed the differential bathymetry image by shifting the bathymetry after the earthquake to reverse the coseismic displacement estimated by the differential bathymetry study (that is, 50 m west-northwest and 10 m down-

wards) and then we calculated the differential bathymetry between the transformed data and the data from before the earthquake (Fig. 2c). If significantly large displacement occurred along the landward-dipping reflectors (that is, the large normal fault at CDP 1,900 and/or the reflectors indicated by the smaller, shallow arrows in Fig. 3), then by this transformation, localized bathymetric anomalies should be visible where those reflectors reach the sea floor. However, the resultant image shows rather uniform results without any clear localizations of slip except at the trench axis (Fig. 2c). These results demonstrate that the overriding plate moved seawards mostly coherently during the earthquake, at least within the resolution of our data. Even after the transformation described above, however, spots of positive anomalies in the seaward half of the frontal prism and an area of negative anomaly at the most landward end of the profile can be observed. This result can be interpreted to indicate that the change in the bathymetry was larger in the most seaward part of the upper-plate material, possibly as a result of elastic shortening or increased fault slip relative to the observed 50 m of slip towards the trench axis. It should be noted that, based on a differential bathymetry image of a profile obtained ~50 km north of the profile examined here, the coseismic slip to the trench axis may extend laterally along the trench, although it seems to become smaller northwards, at least within this 50-km along-trench interval. Moreover, no large positive deviation at the trench axis showing a thrust-up structure in the trench-fill sediment is observed along that northern profile (see Supplementary Information).

As can be clearly recognized in the differential bathymetry image after the transformation (Fig. 2c), the only significant deviation is the pair of negative and positive anomalies at the trench axis (CDP 4,900–5,200 in Fig. 2). Comparison of the seismic sections across the trench, CDP 5,150–5,300, before (Fig. 4a) and after (Fig. 4b) the earthquake shows a remarkable structural difference above

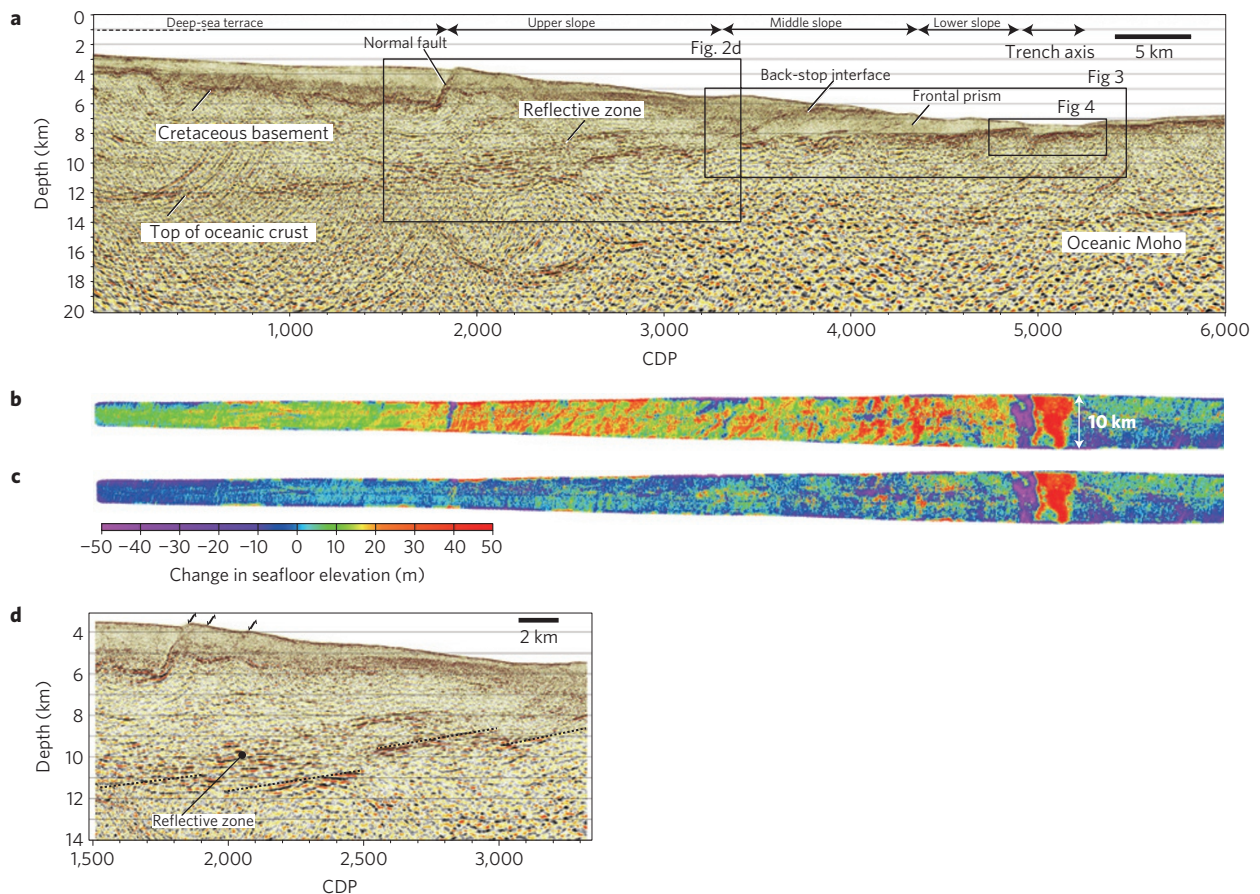


Figure 2 | Seismic reflection and differential bathymetry image seawards of the epicentre. a, Seismic reflection section obtained in 2011. No vertical exaggeration. CDP, common depth point (no unit). **b**, Differential bathymetry image between the bathymetric data acquired in 1999 and the data acquired after the earthquake. **c**, Differential bathymetry image between the 1999 data and the data transformed by shifting the bathymetry obtained after the earthquake horizontally landwards by 50 m and downwards by 10 m. The resolution of the bathymetry data is discussed in detail in ref. 11. **d**, Enlarged seismic section of the reflective zone beneath the upper slope (within the rectangle shown in **a**). Black arrows indicate landward-dipping normal faults.

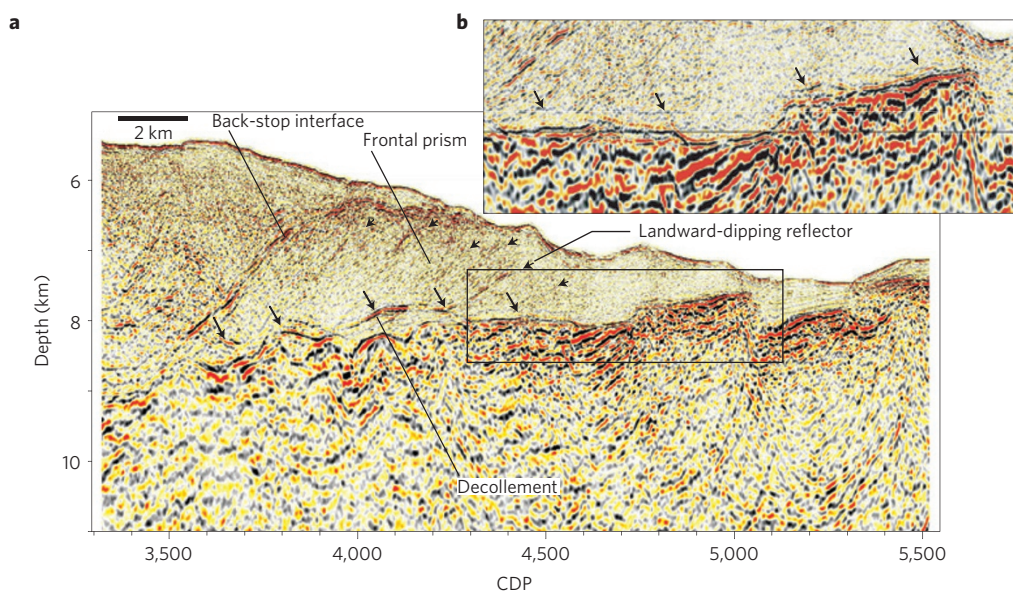


Figure 3 | Seismic image of the frontal prism. a, Enlarged seismic image of the frontal prism (location shown in Fig. 2a) obtained in 2011. Vertical exaggeration, 2:1. **b**, Enlarged section of the area outlined in **a**. Large arrows (**a,b**) indicate a reflector interpreted as the decollement interface. Smaller, shallow arrows in **a** indicate landward-dipping reflectors in the frontal prism.

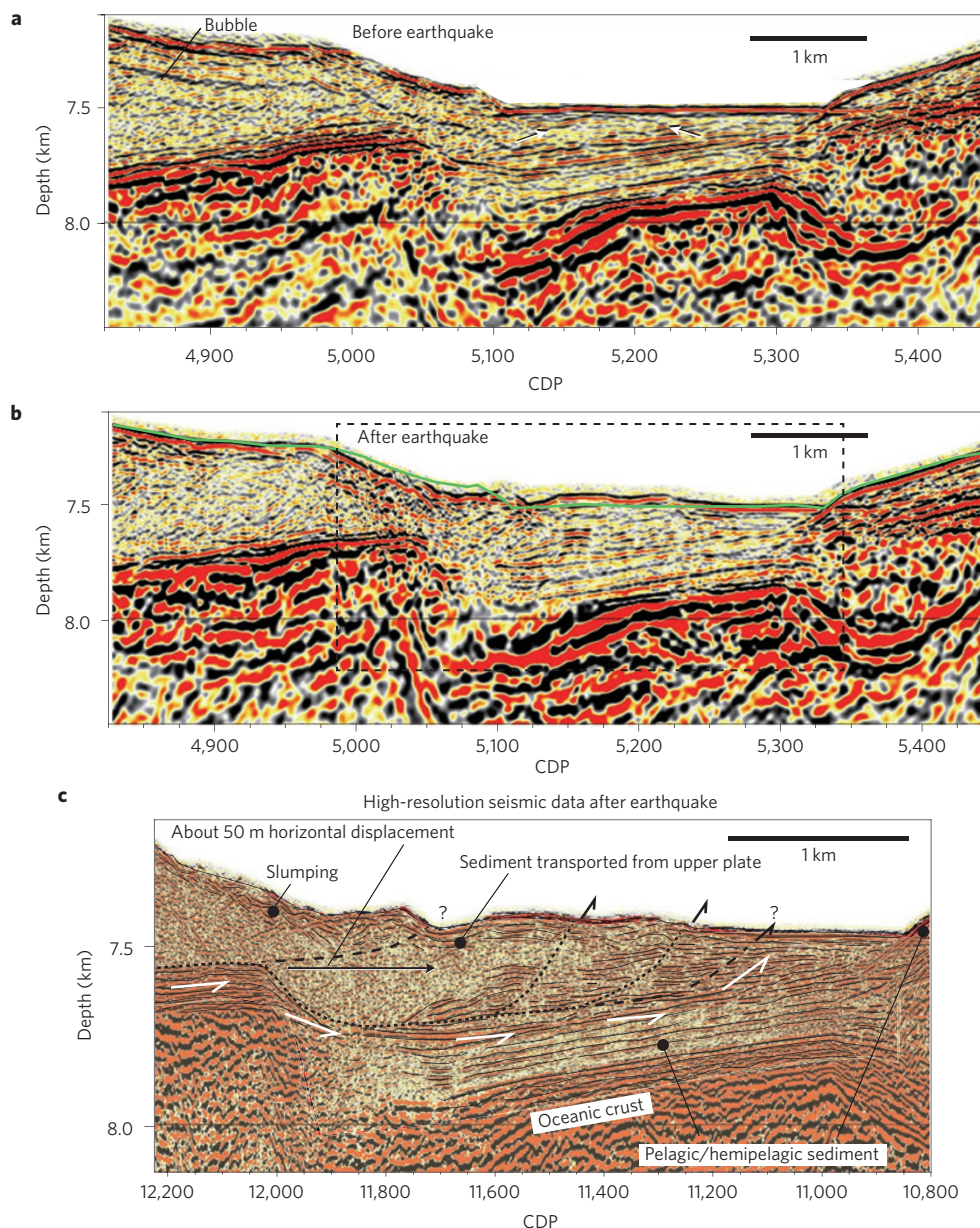


Figure 4 | Comparison of seismic images obtained before and after the earthquake. **a**, Seismic image of the trench axis obtained in 1999 before the earthquake. An interface imaged ~200 m below the sea floor (black interface) at the landward slope of the trench is an artefact generated by a bubble signal from a non-tuned large airgun. Vertical exaggeration, 2:1. **b**, Seismic image of the trench axis obtained after the earthquake. Vertical exaggeration, 2:1. Green line indicates the sea floor before the earthquake. **c**, Interpretation of the high-resolution seismic image obtained after the earthquake around the trench (within the rectangle shown in **b**). See also Supplementary Information.

the well-stratified 150-m-thick sedimentary layer (interpreted as pelagic/hemipelagic sediments because the same sedimentary structure is observed on the incoming plate before subduction²⁴) on the igneous basement. In the seismic section obtained before the earthquake, triangular sediment fill and a flat sea floor are observed at the trench axis, whereas in the seismic section obtained after the earthquake, a 3-km-long, ~350-m-thick deformed/upheaval structure is observed at the trench (Fig. 4c and Supplementary Information). This structure is interpreted as compressional (that is, a thrust-up structure) with reverse faults (Fig. 4c, small black arrows) branching from an interface in the subducting sedimentary layer, which is interpreted as the coseismic master fault (Fig. 4c, white arrows). The seismic image beneath the landward slope of the trench axis, where ~50 m of subsidence is observed in the differential bathymetry data, was imaged after the earthquake as a

seismically scattered structure. This structure is interpreted to have formed by slumping.

We do not know the exact timing of the formation of the deformed structure at the trench axis, but this drastic deformation most probably resulted from coseismic motion during the 2011 earthquake, although we cannot rule out the possibility that postseismic slip occurred after the earthquake but before we acquired the data (on 22 March 2011). It should be noted that the trench-fill sediment may preserve the cumulative displacement by previous earthquakes. A convex-upward interface imaged at the shallowest part in the trench-fill sediment (Fig. 4a, white arrows) could be interpreted as formed by a previous event, as a similar convex-upward structure is observed at the sea floor after the earthquake. However, the age of this structure as well as a record of a fault slip to the trench cannot be determined from the seismic data alone.

Our examination of the observed structure at the trench axis leads us to propose the following scenario for the rupture propagation to the trench. Coseismic slip starting from the epicentre at around 20 km depth²⁵ propagated updip along the reflective zone that is presumably weak, slightly above the subducted igneous oceanic crust. The slip continued to propagate along the interface beneath the frontal prism and slightly above the oceanic crust. The slip finally broke the sea floor at the trench axis (CDP 5,150–5,300, Fig. 4b), creating the associated compressional structure (thrust-up structure) with several reverse faults branching from the master fault in the subducted oceanic sediment. As the overriding block extended over the edge of the horst on the seaward slope, the 50 m of horizontal movement of the overriding block towards the trench brought the sedimentary block at the edge of the horst over the graben, which may have caused gravitational instability, triggering slumping on the seaward slope (Fig. 4c). We emphasize that deformation structures of this type along the Japan Trench axis can be used to identify where a large slip to the toe of the thrust has occurred along the trench, not only during the 2011 earthquake but also during past tsunamigenic earthquakes, in particular to the north, where the Sanriku tsunami earthquake ($M_s = 7.9$; ref. 26) occurred in 1896.

Methods

Data acquisition. For data acquisition during the March 2011 cruise, a tuned airgun array with a total volume of 7,200 cubic inches was used and the seismic waves were recorded by a 444-channel streamer cable ~6,000 m long. The high-resolution seismic data were acquired from 22 October to 11 November around the trench axis by the research vessel *Kaiyo*. The seismic source for the high-resolution seismic survey was a 320-cubic-inch gun array and the receiver was a 1.3-km-long, 192-channel streamer cable with a 6.25-m channel spacing. For data acquisition during the 1999 survey, a seismic source consisting of eight non-tuned airguns, each with a volume of 1,500 cubic inches, was used and the seismic waves were recorded by a 156-channel, ~4,000-m-long streamer cable.

Data processing. We used the same processing sequence, processing parameters and geometry for both the 1999 and the March 2011 cruise data sets. Relative amplitudes were preserved during the processing and conventional data-processing techniques were used, including trace header edit, trace edit, common midpoint (CMP) binning with an interval of 12.5 m, band-pass filter, datum correction, amplitude compensation, minimum phase conversion, predictive deconvolution, velocity analysis, normal moveout correction, mute and CMP stack and Kirchhoff post-stack depth migration. The data-processing flow for the high-resolution data included trace header edit, trace edit, CMP binning with an interval of 3.125 m, band-pass filter, datum correction, velocity analysis, normal moveout correction, mute, CMP stack and post-stack FK time migration and depth conversion using the same velocity applied to the data from the March 2011 cruise.

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Author contributions

S.K. designed the survey, interpreted data and wrote the paper. T.N., Y.N. and S.M. participated in data acquisition and processing. T.F. processed bathymetry data. Yuka K. participated in data acquisition as a chief scientist. N.T., Yoshiyuki K. and A.T. organized data-acquisition cruises. All authors equally contributed to scientific discussions.

Additional information

Supplementary information is available in the online version of the paper. Reprints and permissions information is available online at www.nature.com/reprints. Correspondence and requests for materials should be addressed to S.K.

Competing financial interests

The authors declare no competing financial interests.